

Low-loss Silicon Photonic MEMS Switch Array Based on 2.5D Torus Topology

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Abstract *A scheme for scalable low-loss silicon photonic switch arrays using micro-electro-mechanical systems (MEMS) switches and a novel 2.5D Torus topology is proposed and demonstrated, which exhibits one of the lowest insertion losses among widely used wide-sense and strictly non-blocking topologies. ©2026 The Author(s)*

Introduction

The rapid development of artificial intelligence has driven ever-increasing demands on data processing capabilities, which are related to the scale of computing clusters. Photonic interconnect, with its large capacity and low latency, provides a promising solution to further scale the computing networks, as it is capable of overcoming the intrinsic bottlenecks of conventional electronic interconnect in capacity and latency [1-2]. Central to the high-speed photonic interconnect systems are the reconfigurable silicon photonic integrated switch arrays, which, however, still face challenges in insertion loss and scalability.

To achieve a scalable, low-loss switch array, silicon photonic switches based on various mechanisms have been extensively explored, including electro-optic [3], thermo-optic [4-5], phase-change material [6-7] and MEMS [8-14]. Among these, MEMS switches offer an excellent balance of low loss, low crosstalk, small footprint, and low power consumption, having achieved digitally operated 2×2 elementary switches compatible with standard CMOS fabrication processes [13-14]. These advantages make MEMS a compelling approach for constructing large-scale on-chip switch arrays.

However, the array topology suitable for MEMS switches remains less explored, despite its critical impact on overall performance. Since typical MEMS switches exhibit much lower insertion loss and crosstalk in the OFF state than in the ON state, it is essential to minimize the number of ON-state switches per signal path. While Crossbar [12] and path-independent loss (PILOSS) [15] topology satisfy this requirement, their overall insertion loss remains at a moderate level. For other topologies, Dilated networks [15-16] have a large number of elementary switches,

leading to challenges in packaging. The Benes [17] topology is a rearrangeably non-blocking (RNB) network, and the rearrangement process involves temporary link interruptions, prohibiting its applications in many scenarios such as AI training and inference tasks.

In this work, we propose a novel on-chip 2.5D Torus topology based on the conventional Crossbar for silicon photonic MEMS switch arrays. The proposed topology leverages bypass waveguides to circumvent most switches and waveguide crossings while keeping only a few switches in the ON state. It significantly reduces the path insertion loss to one of the lowest levels among the widely used wide-sense non-blocking (WNB) and strictly non-blocking (SNB) topologies. Meanwhile, it also exhibits relatively low crosstalk and relatively low overall switch count. A 16×16 two-stage 2.5D Torus switch array based on the split waveguide crossing (SWX) MEMS switch [13] is experimentally demonstrated, validating its capability and reliability as a scalable, low-loss switch array scheme.

Principle

The proposed 2.5D Torus topology consists of three components: a split crossbar network (SCN) composed of several $m \times m$ crossbar tiles, N input bypass layers (IBL), and N output bypass layers (OBL), as shown in Fig. 1(a). Each IBL (OBL) includes a $1 \times R$ ($R \times 1$) switch and bypass waveguides to connect those R ports to the $2N/m$ ports in each row (column) of the SCN, where $R = 2N/m$. The schematic of SCN is shown in Fig. 1(b), in which signals are input from the rows and output from the columns. With the IBL (OBL), the input port I_y (output port O_x) can be switched to one of the $2N/m$ ports in the y -th row (x -th column) of the SCN. Therefore, the signal path I_y - O_x can

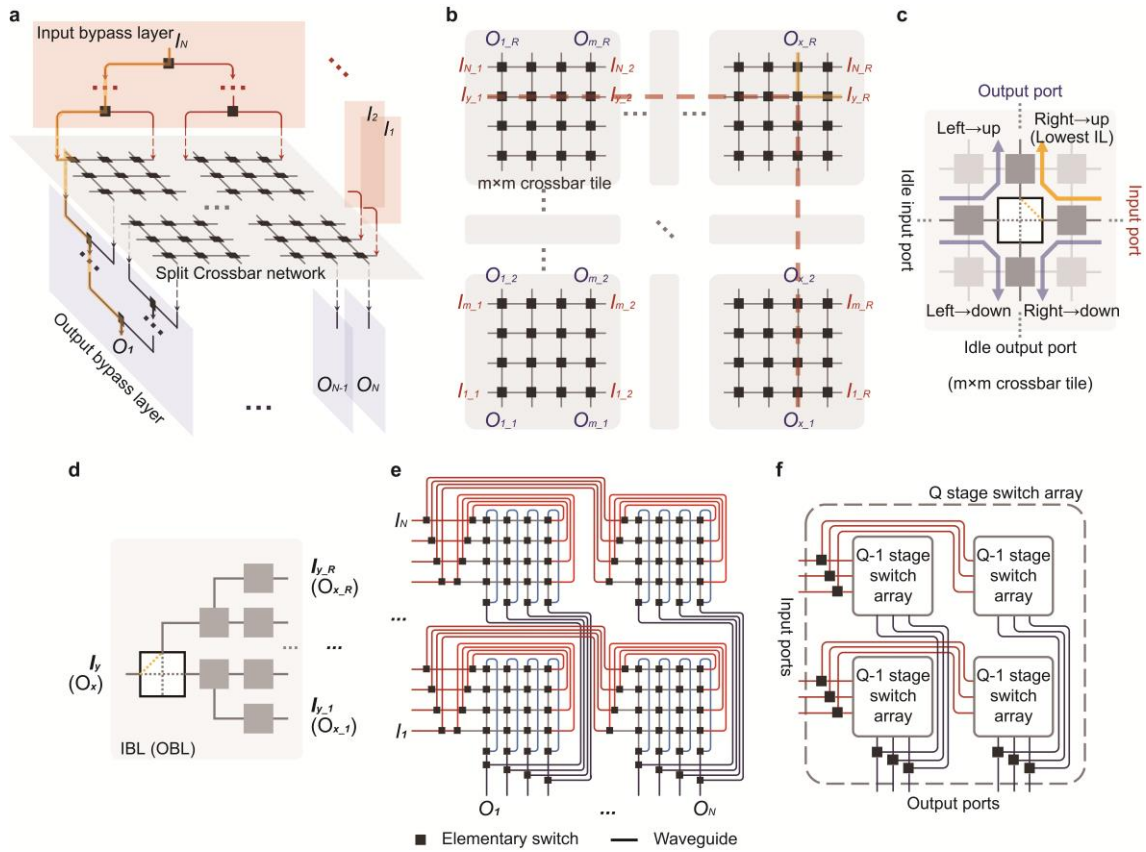


Fig. 1: Construction scheme of the on-chip 2.5D Torus switch array. (a) Schematic of the 2.5D Torus topology. (b) Schematic of the SCN. (c) Switch configuration at the crosspoint of SCN. (d) Switch configuration in the IBLs and OBLs. (e) Schematic of a two-stage planar 2.5D Torus switch array. (f) Method to scale the switch array.

circumvent all the switches in other crossbar tiles of the SCN, except those in the tile containing the crosspoint (x, y) , thereby significantly reducing path insertion loss.

The proposed 2.5D Torus topology can be constructed using typical 1×2 elementary switches. In the SCN, the 1×2 elementary switch at each crosspoint (x, y) has four possible switch directions (left to up, left to down, right to up, right to down), corresponding to four possible routes to form signal path $I_y \rightarrow O_x$. This elementary switch should be configured to the direction that enables the route with the lowest insertion loss, as shown in Fig. 1(c). All the elementary switches in the SCN can be configured using this method. In the IBLs and OBLs, the $1 \times R$ or $R \times 1$ switch can be cascaded from 1×2 elementary switches, as shown in Fig. 1(d).

Fig. 1(e) and 1(f) provide a solution to merge all the bypass layers and the SCN into a planar 2.5D Torus topology, which can be directly applied to an on-chip switch array. First, four crossbar tiles are used to construct a two-stage planar 2.5D Torus array as shown in Fig. 1(e). Then four two-stage arrays are used to construct a three-stage array as shown in Fig. 1(f). Continuous scaling can be achieved by repeating this step.

The proposed on-chip 2.5D Torus topology exhibits low insertion loss, relatively low crosstalk,

and relatively low overall switch count. Using the typical insertion loss values of a MEMS switch (0.2 dB for OFF state, 0.5 dB for ON state) and a waveguide crossing (30 mdB) [13], as shown in Fig. 2(a) and 2(b), the estimated maximum path insertion loss of the 2.5D Torus array based on 8×8 crossbar tiles is one of the lowest among the widely used WNB and SNB topologies (Fig. 2(c)). In addition, undesired first-order crosstalk can only occur between a subset of signal paths within a single crossbar tile, while other signal paths experience at worst second-order crosstalk, offering an advantage over Crossbar and PILOSS topologies. Furthermore, the overall

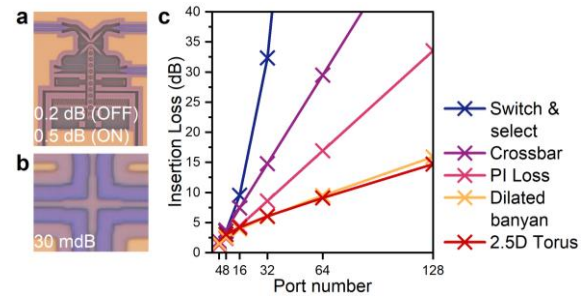


Fig. 2: Estimated on-chip insertion loss based on typical elementary switches and waveguide crossings. (a) SWX MEMS switch. (b) Varied-width multimode interference waveguide crossing. (c) Comparison of maximum path insertion loss among different topologies.

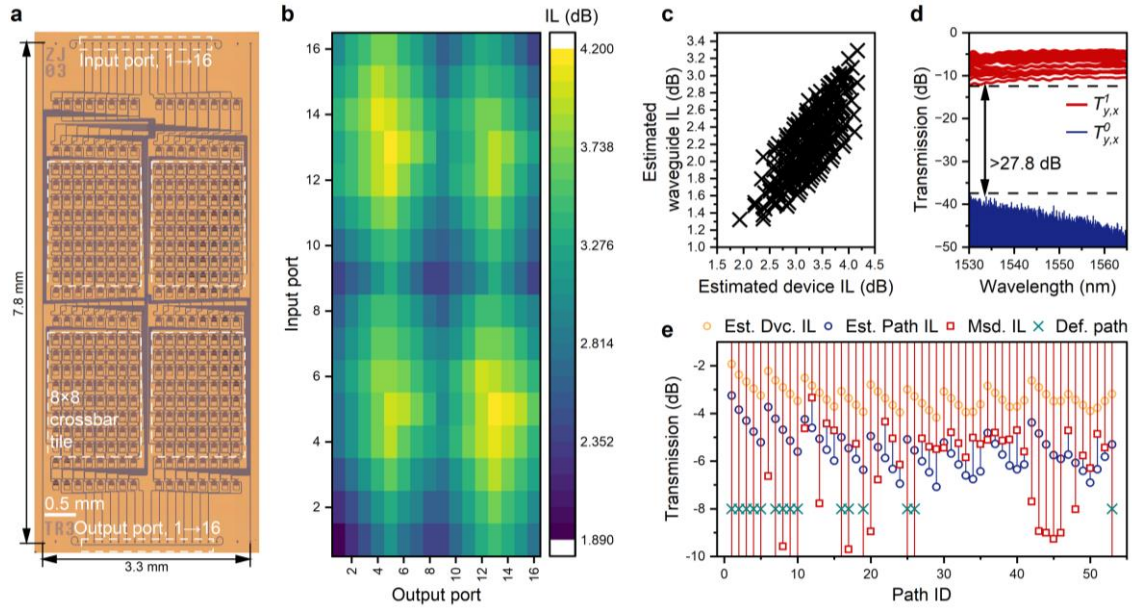


Fig. 3: Estimation and experimental results of the 16×16 two-stage 2.5D Torus MEMS switch array. (a) Optical microscope image of the switch array. (b) Estimated insertion loss of each signal path induced by devices. (c) Estimated waveguide loss and device loss in each signal path, respectively. (d) Measured transmission spectra of the 53 tested signal paths. (e) Measured and estimated insertion loss of the 53 tested signal paths at 1550 nm. Estimated: Est.; Device: Dvc.; Measured: Msd.; Defective: Def.

switch count in such an array is $\frac{4-m}{m}N^2 - 2N$, which lies between that of Crossbar and PILOSS (N^2) and that of Dilated Banyan and Switch & Select ($2N^2 - 2N$).

Experiment

Based on the 2×2 SWX MEMS switch [13], a planar 16×16 two-stage 2.5D Torus switch array is designed and fabricated (refer to [13] for fabrication process). The switch array consisting of 352 elementary switches has an overall footprint of 7.8×3.3 mm, as shown in Fig. 3(a).

The total on-chip insertion loss of signal paths consists of two parts: device loss from switches and waveguide crossings, and waveguide propagation loss. The estimated device loss ranges from 1.9 to 4.2 dB, as shown in Fig. 3(b). However, due to the high propagation loss of the fabricated waveguides (8.3 dB/cm for the 600-nm ridge waveguide), the total waveguide loss is non-negligible, amounting to 1.3–3.3 dB, as shown in Fig. 3(c), which consequently degrades the on-chip path insertion loss to 3.2–7.5 dB. By employing a standard silicon photonic foundry process, the waveguide propagation loss can be reduced to 2 dB/cm for the 450-nm strip waveguide, significantly reducing the total waveguide loss to about 0.4–1.1 dB.

The fabricated switch array was characterised with four electrical probes and a pair of optical fibers. First, for each signal path I_y-O_x , two or three switches were switched on to form the route, and the transmission spectrum $T^1_{y,x}$ was measured. Then, all the switches were switched off, and the

transmission spectrum $T^0_{y,x}$ was measured at the same ports. The transmission spectra of 53 tested signal paths were measured in this procedure, as shown in Fig. 3(d), exhibiting an extinction ratio over 27.8 dB ($T^1_{y,x} - T^0_{y,x}$) in the C band. The on-chip insertion loss of those signal paths at 1550 nm is 3.3–9.3 dB, denoted by the red squares in Fig. 3(e), excluding the defective paths denoted by grey crosses. For two-thirds of the tested signal paths, the difference between measured loss and estimated loss (denoted by the blue circles, 3.3–7.5 dB) is acceptable. The defective paths are mainly attributed to electron-beam lithography (EBL) stitching errors, slight fabrication defects, switch performance degradation induced by poor metal-silicon contacts, and random particulate contamination on the chip surface.

Conclusion

In this work we propose and demonstrate a novel 2.5D Torus topology for silicon photonic MEMS switch arrays, which exhibits one of the lowest levels of insertion loss among widely used WNB and SNB topologies, while achieving low cross-talk, low overall switch count, and compatibility with typical 1×2 MEMS switches. A 16×16 two-stage 2.5D Torus MEMS array was also experimentally demonstrated, with the measured on-chip insertion loss of 53 tested signal paths ranging from 3.3 to 9.3 dB. Our scheme provides a promising solution for large-scale, low-loss on-chip switch arrays and further for photonic interconnect and switching in data centers.

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